

* Some Applications of the First order D.E: ①

Growth Low:

P_0 : The Initial Population at time $[t=0]$

t : time

$P(t)$: The Population at any time

The rate of Population ^{amount} is proportional to the Population at any time

$$\begin{array}{l} \frac{dP(t)}{dt} \propto P(t) \\ \text{proportional relation} \\ P(0) = P_0 \text{ Initial condition} \end{array}$$

$$\rightarrow \frac{dP(t)}{dt} = k P(t) \quad \text{--- (1)}$$

The proportional constant

From equ(1) $\Rightarrow \frac{dP(t)}{P(t)} = k dt$ (separate variables) --- (2)

$$\int \frac{1}{P(t)} dP = \int k dt = \ln(P(t)) = kt + C_1$$

Integrate both sides of equ(2) $\Rightarrow P(t) = \frac{e^{kt+C_1}}{e^{kt+C_1}} = e^{kt} \cdot e^{C_1}$

$P(t) = C e^{kt}$ general solution

$$P(0) = C e^{k(0)}$$

$$P(0) = C e^0 = P_0 \Rightarrow C = P_0$$

$$\therefore P(t) = P_0 e^{kt} \quad (\text{Formula})$$

EX#1) The population rate of a bacteria culture is (2)
 Proportional to the population at any time. Suppose
 Initially we have $10,000$ bacteria and after (2) hours
 we found the population of the bacteria is $12,000$ units $\rightarrow k = ??$
 How many hours will take to have $20,000$ units?

Solution, $P'(t) = kP(t)$, $P(0) = 10,000 = P_0$
 $P(2) = 12,000$

① $P(t) = P_0 e^{kt} \Rightarrow P(t) = 10,000 e^{kt} \quad \text{--- (1)}$

② $P(2) = 10,000 e^{2k}$

$12,000 = 10,000 e^{2k} \Rightarrow e^{2k} = \frac{12,000}{10,000} = 1.2$

Take $\ln$ to both sides:

$2k = \ln(1.2) \Rightarrow k = \frac{\ln(1.2)}{2} = 0.09$

$\therefore P(t) = 10,000 e^{0.09t} \quad \text{--- (2)}$

③ $20,000 = 10,000 e^{0.09t}$
 $e^{0.09t} = 2$

$0.09t = \ln(2)$

$t = \frac{\ln(2)}{0.09}$

$t = 7.7 \text{ hours}$

Ex #2) The population of a community is known to increase at a rate proportional to the number of people present at a time t . If the population has doubled in 6 years, how long it will take to triple?
Find t ?

Solution) $\frac{dp}{dt} = k p(t)$

Find t where $p(t) = 3p_0$? $\underbrace{p(6) = 2p_0}_{\text{calculated previously}}, p(0) = p_0$

$p(t) = p_0 e^{kt}$
calculated previously

$$2p_0 = p_0 e^{6k} \Rightarrow 2 = e^{6k} \Rightarrow 6k = \ln(2) \Rightarrow k = \frac{\ln(2)}{6}$$

$$\Rightarrow k = \frac{0.693}{6} = \boxed{0.1155}$$

$p(t) = p_0 e^{\overbrace{\left(\frac{\ln(2)}{6}\right)t}^{0.1155}} = 3p_0$

$$0.1155 t = \ln(3) \Rightarrow t = \frac{\ln(3)}{0.1155} = \frac{1.0986}{0.1155} = \boxed{9.5117} \text{ Years}$$

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#3) The rate of increase of a colony of bacteria is (4) proportional to the number of bacteria present at any particular time. If initial number of bacteria is 1000 and (4) hours later there are 1350. After how many hours the population will be 2000 units?

Solution) $P_0 = 1000 \Rightarrow P(4) = 1350$ (This is given to find K)

$$P(t) = P_0 e^{kt}$$

$$P(4) = 1350 = 1000 e^{4k}$$

$$\Rightarrow \frac{135}{100} = e^{4k} \Rightarrow \ln(1.35) = 4k \Rightarrow k = \frac{\ln(1.35)}{4}$$

$$\Rightarrow k = \frac{0.3}{4} = 0.075$$

$$P(t) = 1000 e^{0.075t} \quad (*)$$

To find t?

$$2000 = 1000 e^{0.075t}$$

$$2 = e^{0.075t}$$

$$\ln(2) = 0.075t$$

$$t = \frac{\ln(2)}{0.075}$$

$$t = 9.24 \text{ hours}$$

Ex #4] One hundred Fruit flies are placed in breeding container that can support a population of at most 5000 flies. If the population grows at a rate of 2% per day, how long will it take for the container to reach capacity.
 $P_0 = 100$
 $K = 0.02$
 ↳ maximum

Solution) $k = 2\% = 0.02$

Find t , where $P(t) = 5000$. Note that $P(t) = P_0 e^{kt}$

$$100 e^{0.02t} = 5000 \quad (\text{Divide by 100 both sides})$$

$$e^{0.02t} = 50 \quad (\text{Take Ln to both sides})$$

$$0.02t = \ln(50)$$

$$t = \frac{\ln(50)}{0.02} = 195.6 \text{ days}$$

Ex #5] In a certain chemical reaction a substance decomposes at a rate proportional to the amount present. Tests show that 15000 grams will reduce to 5000 grams in 10 hours.

(a) Find a formula for the amount that will remain from a 15000 grams sample after t hours.

Solution) $\frac{dP}{dt} = kP(t)$, $P(0) = 15000$

$$P(10) = 5000 \text{ grams}$$

$$P(t) = P_0 e^{kt} \Rightarrow P(t) = 15000 e^{kt}$$

$$P(10) = 15000 e^{10k} = 5000 \Rightarrow e^{10k} = \frac{5}{15} = \frac{1}{3}$$

Take $\ln$ to both sides:

$$10k = \ln\left(\frac{1}{3}\right) = \ln(1) - \ln(3) = -\ln(3)$$

$$k = \frac{-\ln(3)}{10} \approx \boxed{-0.11} \text{ b/c } k \text{ decrease}$$

$$\therefore \boxed{P(t) = 15000 e^{-0.11t}}$$

⑥ How Long it take for 50% of an initial sample of $P(0)$ grams to decompose? $\frac{50}{100} = \boxed{\frac{1}{2}}$

i.e Find t where $P(t) = \frac{1}{2}(15000) = 7500$
50% decay

$$15000 e^{-0.11t} = 7500$$

$$e^{-0.11t} = 0.5$$

$$-0.11t = \ln\left(\frac{1}{2}\right) = \ln(1) - \ln(2)$$

$$-0.11t = -\ln(2)$$

$$t = \frac{\ln(2)}{0.11} \approx 6.3 \text{ (Hours)}$$

Ex. #6) (Simple chemical processes)

⑦

Suppose that a dose of d mg of a drug is injected into the blood stream. Assume that the drug leaves the blood and enters the urine at a rate proportional to the amount of the drug present in the blood. Assume in addition that half of the drug dose has entered the urine after one ^{(E=1) hour} hour. Find the time at which the amount of drug in the blood stream is 20% of the original dose d .

$$\frac{20}{100} = \frac{2}{10} = 0.2 \text{ or } \frac{1}{5}$$

Solution) $P(0) = d = P_0$

$$P(t) = P_0 e^{kt}$$

$$P(t) = d e^{kt}$$

$$P_0 = d$$

$$P(1) = \frac{1}{2} d$$

$$\frac{1}{2} d = P(1) = d e^{k(1)} \Rightarrow 0.5 = e^k \Rightarrow k = \ln\left(\frac{1}{2}\right) = -\ln(2)$$

↳ delay

$$\therefore P(t) = d e^{-\ln(2)t}$$

We need to find t , when $P(t) = 0.2d$.

$$0.2d = d e^{-\ln(2)t}$$

$$0.2 = e^{-\ln(2)t}$$

$$0.2 = e^{-(0.693)t} \quad \text{Take } \ln \text{ to both sides:}$$

$$t = \frac{\ln(0.2)}{-0.693}$$

$$t = 2.32 \text{ Hours}$$